

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	T-12
Project Name	FinVault – Virtual Stock Trading Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Account Registration	• Custom signup form submission (name, email, password) • Profile picture upload and directory handling via Multer middleware • Secure password encryption using bcrypt before database storage
FR-2	User Account Authentication	• Credential validation against hashed password values • JSON Web Token (JWT) generation and localStorage placement on client • API request interception to attach authorization Bearer headers
FR-3	Stock Catalog & Search	• Display real-time list of listed stocks (symbol, company, price) • Query filters for stock searches by company symbol or name • Interactive stock details page with historical price graphs using Chart.js
FR-4	Personal Watchlist	• Enable users to pin specific stocks for regular tracking • Dynamically add or remove stocks from user watchlist database records
FR-5	Virtual Trading & Portfolio	• Wallet balance tracking (initialized with \$100,000 mock money) • Buy Order processing: verifies balance, updates holdings, logs ledger • Sell Order processing: verifies ownership, updates balance, logs ledger • Live calculations of portfolio value, average buy price, and net returns
FR-6	Administrative Controls	• Admin dashboard overview of total users, orders, and stock count • Manage user directories (view records, monitor trading actions) • Stock catalog management (add, edit, update prices, delete stocks)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The platform must provide a responsive user interface that scales seamlessly from desktops to mobile devices, featuring modern, dark-themed styling, clear data charts with tooltips, and real-time user validation alerts.
NFR-2	Security	All user passwords must be hashed using salt rounds (bcrypt). Client session states must be protected using stateless JWT tokens, and CORS policies must restrict API access to whitelisted domains only.
NFR-3	Reliability	Trading actions (updating wallet balance and portfolio stock quantities) must occur inside atomic database operations to prevent data discrepancies. DB connection retry handlers must handle temporary network dropouts.
NFR-4	Performance	Stock listing and search results should load in under 100ms. Search inputs must be debounced to restrict redundant API requests, and MongoDB indexes should be maintained on heavily queried columns like `symbol` and `email`.
NFR-5	Availability	The database layer must ensure 99.9% uptime by leveraging a 3-node MongoDB Atlas replica set with automated failover. The Express API server must be deployed on cloud containers with automatic health-checks.
NFR-6	Scalability	Decoupled 3-tier MERN architecture must allow the React SPA (hosted on Edge CDNs) and stateless Express API services to scale independently based on user traffic and requests per second.